Data Set Report

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1 Introduction

This report outlines the process used to expand the real estate historical dataset. The objective was to create a comprehensive price history for multiple societies in Islamabad and Rawalpindi, covering a wide range of plot sizes (4 to 20 Marla) over a timeline from 2020 to 2026.

2 Data Sources

The data generation process utilized two sources:

- **Base Dataset:** New data was generated for DHA Islamabad, Bahria Town, Capital Smart City, Faisal Town, Faisal Hills, and Grace Valley. These datasets were created initially for 5 Marla and 10 Marla sizes, following realistic market trends relative to the base dataset, before being fed into the extension algorithm.

3 Python Extension Logic

The Python script was designed to ensure data consistency and linear scalability. The process involved the following key steps:

3.1 Rate Normalization

To prevent conflicts where multiple sizes (e.g., 5 Marla and 10 Marla) existed for the same date and society, the script grouped the data by unique events. It calculated the **Mean Price Per Marla** for each specific combination of Date, Society, Block, and Development Status. This established a single, authoritative "Market Rate" for that specific time and location.

3.2 Linear Scaling (The Multiplication Process)

The script utilized the existing `Price_Per_Marla_PKR` value as the primary driver for expansion. For every historical entry, the script generated new rows for the following sizes:

$$[4, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] \text{ Marla}$$

The calculation logic applied was:

$$\text{New Price} = \text{Price\_Per\_Marla\_PKR} \times \text{New Size}$$

Example: If the existing rate for Faisal Town Block A in Jan 2020 was 580,000 PKR/Marla, the script generated a price for 20 Marla by calculating:

$$580{,}000 \times 20 = 11{,}600{,}000 \text{ PKR}$$

3.3 Data Merging

The newly generated rows were assigned sequential IDs and appended to the original dataset. The final output was a single, unified CSV file containing all societies and all requested sizes.

4 Conclusion

The resulting dataset provides a granular view of property prices across six distinct societies. By using the `Price_Per_Marla_PKR` as the multiplication factor, the data maintains logical pricing consistency across different plot sizes, reflecting a linear relationship between plot area and total market value.